

Conformal Prediction for Hierarchical Data

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Motivation and Contributions

Many real-world regression tasks involve **multivariate targets** that follow a so called **hierarchical structure**. Such structural information can be leveraged in a post-hoc manner for point forecasting by **Forecast Reconciliation** (FR) but only few approaches exist for uncertainty quantification.

In the article, our contributions are:

- combining FR with **Conformal Prediction** (CP)
- showing that the **validity** granted by SCP remains
- showing that it improves the **efficiency** of the conformal sets

Hierarchical Data

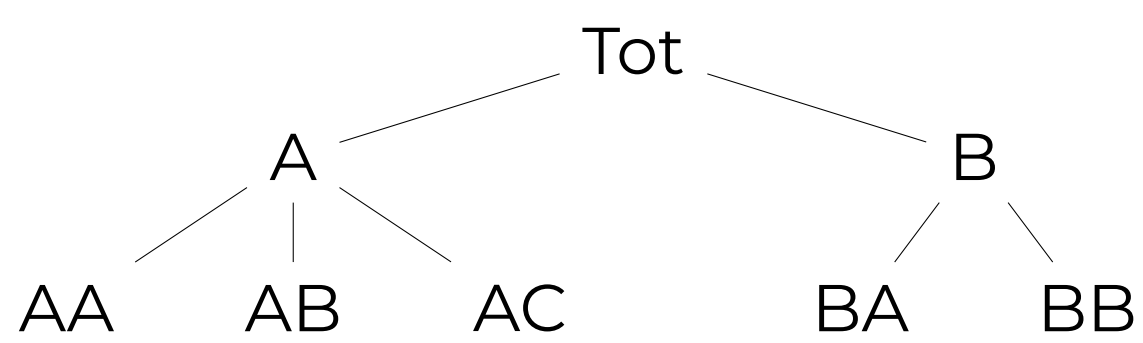
- $\mathbf{y}_t \in \mathbb{R}^m$, the target and $\mathbf{x}_t \in \mathbb{R}^d$, the features at time t
- $\mathbf{y}_{t,1:n} \in \mathbb{R}^n$, the most disaggregated level of the target $\mathbf{y}_t$
- $H \in \mathbb{R}^{m \times n}$, the structural matrix (usually noted S)
- $(\mathbf{x}_t, \mathbf{y}_t)_{t \in \llbracket 1, T \rrbracket}$, available observations split between $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{calib}}$

The hierarchical structure is then defined by the constraint:

$$\mathbf{y}_t = H\mathbf{y}_{t,1:n} \quad (1)$$

Example: a typical hierarchical structure

$$H = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ & & & \text{Id}_5 \end{pmatrix}$$



A real-world example is household expenditure, where spending on food, housing, taxes, etc. sums to the total expenditure.

Forecast Reconciliation

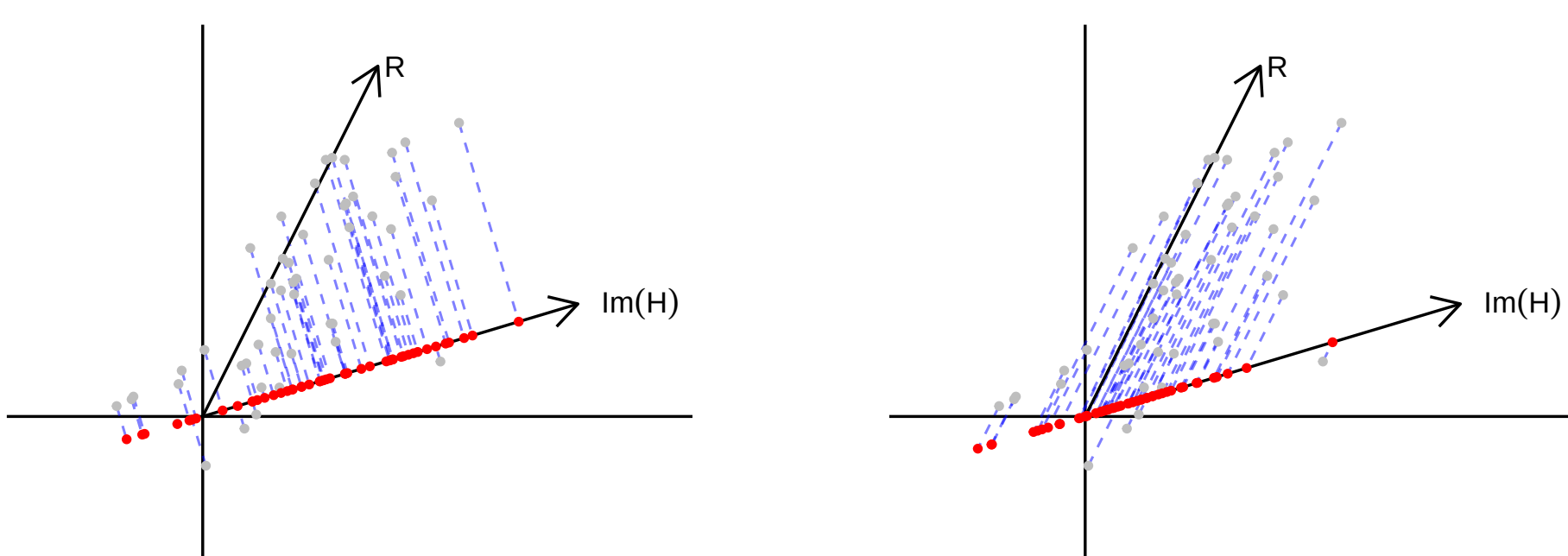
FR is a **post-hoc** procedure based on the idea that if the target $\mathbf{y}_t$ follows a hierarchical structure, then its forecast $\hat{\mathbf{y}}_t$ should follow the same linear constraints (1).

In the article, we concentrate on FR through projection, which is an intuitive and efficient approach.

Reconciliation via Projections [Panagiotelis et al, 2021]

$$\tilde{\mathbf{y}}_t = HG \cdot \hat{\mathbf{y}}_t \quad \text{s.t.} \quad GH = \text{Id}_n \quad (2)$$

How to choose the projection matrix?



For robust results, we can consider the **orthogonal projection** (OLS) but we can also reduce the variance of the forecast errors with a **data-based projection** (MinT).

Conformal Prediction

In the article, we target either joint or component-wise coverage. Here we focus on the **component-wise objective**.

The benchmark approach (Direct) consists in running independently SCP for each component. In particular, we consider signed non-conformity scores $\hat{s}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ and the associated conformal prediction procedure.

Reconciled Conformal Prediction

To leverage the hierarchical structure in SCP, we now include a **reconciliation step** in the computation of the non-conformity scores i.e. we consider $\tilde{s}_t = P \cdot \hat{s}_t$ instead of $\hat{s}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t$ where P is a projection onto $\text{Im}(H)$.

Theorem 1. Reconciled conformal prediction, used with any matrix P such that $PH = H$, ensures that, whenever the scores are i.i.d. and almost surely distinct

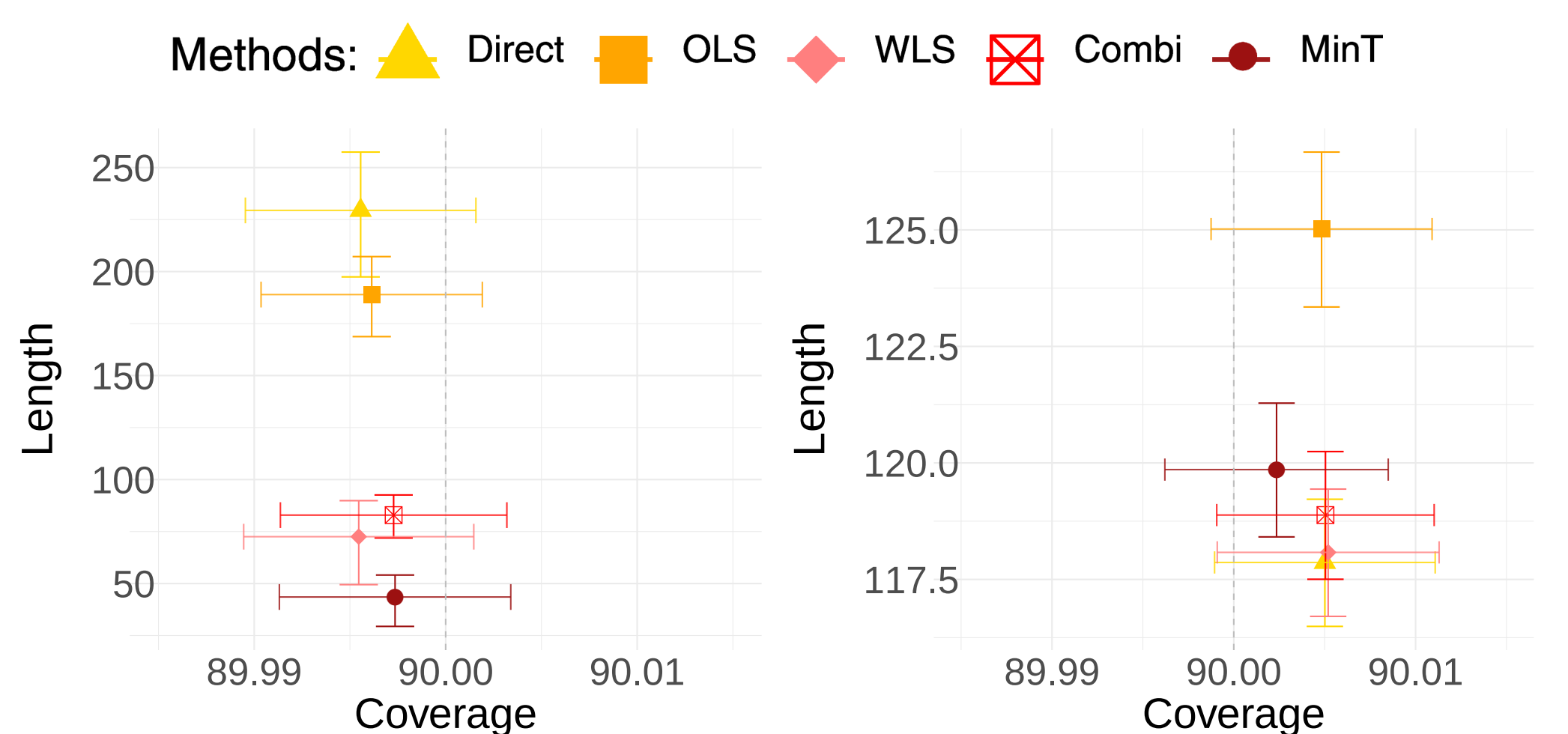
$$\forall i \in \llbracket 1, m \rrbracket, \quad \mathbb{P}(y_{T+1,i} \in \tilde{C}_i(x_{T+1})) \approx 1 - \alpha.$$

We only display the strongest efficiency result of the article. The pitfall of this result is that it requires the knowledge of the covariance matrix Σ of the forecast errors.

Theorem 2. Suppose the scores are i.i.d. and elliptical. Then, the oblique projection in Σ^{-1} -norm (MinT) is the optimal reconciliation step and achieves

$$\forall i \in \llbracket 1, m \rrbracket, \quad \mathbb{E}[\mathcal{L}_1(\tilde{C}_i^*(x_{T+1}))^2] \leq \mathbb{E}[\mathcal{L}_1(\tilde{C}_i(x_{T+1}))^2].$$

Numerical Illustrations



Config.	Direct	OLS	WLS	Combi	MinT
1	876 ± 254	787 ± 226	322 ± 131	364 ± 101	216 ± 47
2	871 ± 253	753 ± 216	308 ± 116	361 ± 92	246 ± 51
3	3032 ± 467	2954 ± 455	1869 ± 377	1758 ± 395	1502 ± 578
4	3036 ± 479	2901 ± 458	1581 ± 340	1604 ± 349	1404 ± 571
5	10424 ± 885	10358 ± 880	8861 ± 853	9664 ± 850	10613 ± 918
6	10621 ± 889	10460 ± 875	7673 ± 785	9068 ± 806	10503 ± 905

Table: Square-root empirical averages of $\|\mathcal{L}_1(\tilde{C}(x_{T+1}))\|^2$.

References

- A. Panagiotelis, G. Athanasopoulos, P. Gamakumara, and R. Hyndman. Forecast reconciliation: A geometric view with new insights on bias correction. *International Journal of Forecasting*, 37(1):343–359, 2021.